

GABORONE, BotswanaWMO: **682400**Lat: **24.555S**Lon: **25.918E**Elev: **3299**StdP: **13.03**Time Zone: **2.00 (E02)**Period: **95-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.4	37.8	6.6	8.7	64.5	10.2	10.5	63.3	20.2	64.9	17.9	65.8	4.0	270	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.5	98.4	62.5	95.8	62.6	93.5	62.7	71.0	79.4	70.2	78.8	69.3	78.5	9.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	123.2	72.4	68.2	117.3	71.9	67.6	114.6	71.9	37.1	79.5	36.4	77.8	35.6	79.2	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	17.5	15.4	DB	29.1	102.0	3.3	2.6	26.7	103.8	24.7	105.3	22.8	106.8	20.4	108.6	(4)
(5)			WB	24.0	73.0	2.5	0.6	22.2	73.4	20.7	73.8	19.3	74.1	17.4	74.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.6	78.2	78.0	74.7	68.6	61.7	56.8	56.0	62.1	69.8	75.6	76.9	77.8	(6)	
	DBStd	9.39	4.25	4.21	4.16	4.43	4.49	3.91	4.15	5.27	5.46	5.48	5.34	4.67	(7)	
	HDD50	10	0	0	0	0	1	3	5	1	0	0	0	0	(8)	
	HDD65	805	0	0	1	16	119	247	278	121	19	2	2	0	(9)	
	CDD50	7177	874	783	767	559	365	207	192	376	593	794	807	860	(10)	
	CDD65	2496	409	363	303	125	17	1	0	31	162	331	358	396	(11)	
	CDH74	27196	4009	3597	2723	1180	542	137	145	738	2230	3945	3926	4024	(12)	
	CDH80	12131	1876	1691	1131	344	97	11	10	185	971	1969	1945	1901	(13)	
(14) Wind	WSAvg	7.0	7.4	6.9	6.3	5.8	5.3	5.4	5.8	7.1	8.2	9.1	8.6	7.9	(14)	
(15) Precipitation	PrecAvg	17.0	3.1	2.6	2.4	0.9	0.5	0.3	0.0	0.0	0.2	1.3	2.5	2.8	(15)	
	PrecMax	33.0	7.9	9.5	6.6	5.4	3.6	3.8	0.4	0.2	2.5	4.3	10.9	8.1	(16)	
	PrecMin	8.4	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.9	(17)	
	PrecStd	6.1	2.4	2.6	1.9	1.2	0.7	0.7	0.1	0.1	0.5	1.1	2.1	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.3	98.8	97.1	91.0	87.4	81.0	81.0	88.2	95.3	100.0	100.7	99.0	(19)	
		MCWB	63.4	63.9	62.3	60.9	56.7	53.6	52.7	56.9	59.1	60.8	62.1	63.9	(20)	
	2%	DB	95.4	95.3	93.1	87.5	82.5	77.4	77.6	84.8	91.9	96.7	96.9	95.9	(21)	
		MCWB	64.4	64.5	64.0	61.9	56.0	53.0	51.4	55.5	57.9	60.2	61.6	64.0	(22)	
	5%	DB	93.0	92.8	89.6	84.2	79.7	75.1	75.0	82.0	89.6	93.4	93.5	93.0	(23)	
		MCWB	64.8	65.0	63.7	61.2	55.5	52.2	50.9	54.3	57.9	59.9	62.5	64.5	(24)	
	10%	DB	89.6	89.5	86.2	80.9	77.0	71.8	71.6	78.5	86.2	90.0	91.0	89.7	(25)	
		MCWB	65.4	65.5	64.2	61.2	55.4	52.1	51.2	53.6	56.2	59.9	61.7	65.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	71.6	71.1	68.2	62.1	58.0	56.5	60.0	63.3	66.7	69.5	71.4	(27)	
		MCDB	80.2	78.7	79.5	75.6	74.7	71.8	72.0	81.6	83.3	82.8	79.7	81.1	(28)	
	2%	WB	71.1	70.6	70.1	66.6	60.7	56.2	54.8	58.3	61.8	65.5	68.1	70.2	(29)	
		MCDB	79.2	78.6	78.2	76.4	74.2	70.6	70.6	79.4	81.9	82.1	79.6	80.2	(30)	
	5%	WB	70.1	69.7	69.2	65.6	59.3	54.7	53.4	57.1	60.6	64.3	67.1	69.2	(31)	
		MCDB	78.5	78.6	76.9	76.0	73.7	70.1	69.2	77.5	81.4	80.8	79.9	79.6	(32)	
	10%	WB	69.2	68.9	68.2	64.4	57.6	53.2	52.0	55.6	59.3	63.2	66.1	68.2	(33)	
		MCDB	78.7	78.6	75.7	74.7	72.5	69.7	69.0	74.7	80.7	80.6	80.2	79.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.5	20.9	21.7	23.6	30.1	31.2	32.1	31.7	29.8	26.2	23.3	21.3	(35)	
		MCDBR	26.1	26.4	27.6	28.5	34.2	36.4	36.8	35.8	34.3	31.1	28.3	26.2	(36)	
		MCWBR	6.9	7.7	8.1	9.2	14.3	16.6	17.6	14.7	13.5	11.0	8.8	7.0	(37)	
	5% WB	MCDBR	18.5	18.8	18.8	20.6	27.3	29.6	30.4	30.4	28.0	25.3	22.4	19.7	(38)	
		MCWBR	6.0	6.4	6.8	7.6	11.9	14.4	15.7	12.8	12.4	9.9	7.9	6.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.382	0.389	0.372	0.344	0.302	0.287	0.286	0.348	0.397	0.421	0.372	0.375	(40)	
	taud		2.389	2.364	2.396	2.433	2.513	2.525	2.512	2.333	2.202	2.186	2.371	2.404	(41)	
	Ebn at Noon		305	298	294	289	290	288	292	283	282	285	306	308	(42)	
	Edh at Noon		41	41	38	34	28	26	28	36	45	49	42	41	(43)	
(44) All-Sky Solar Radiation	RadAvg		2205	2040	1847	1577	1448	1296	1407	1659	1962	2141	2221	2227	(44)	
	RadStd		215	193	170	143	61	67	41	73	84	110	210	151	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page